RANBIR KUMAR

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PROFESSIONAL SUMMARY

AI & ML undergraduate (CGPA 8.28) building production deep learning and computer vision systems with PyTorch, TensorFlow, and YOLOv8 on edge hardware (Jetson Xavier NX, Raspberry Pi), including <40 ms real-time inference in industry deployments. Experienced in RAG/semantic retrieval (FastAPI, FAISS, pgvector, LangChain/LlamaIndex) and explainable AI (SHAP, LIME). Filed patent, peer-reviewed research in vision-based perception and few-shot medical AI, and rover autonomy team lead (20th worldwide, IRC 2025). Targeting AI Engineer / Deep Learning roles.

EDUCATION

B.Tech, Computer Science (AI & ML) — Vellore Institute of Technology, Chennai **| CGPA: 8.28/10 |** Aug 2023 – May 2027 (expected)
Relevant Coursework: Machine Learning, Deep Learning, Reinforcement Learning, Explainable AI, Speech & NLP, Computer Vision, Machine Vision, Data Structures & Algorithms, Probability & Statistics, SQL
Certifications: Getting Started with Artificial Intelligence — IBM SkillsBuild · Journey to Cloud: Envisioning Your Solution — IBM SkillsBuild

EXPERIENCE

Technical Integration Lead | IORA, Noida | May 2026 – Present
- Own technical integration for an nRF9151-based LTE-M/NB-IoT safety wearable; drive architecture reviews across PCB, firmware (MQTT over TLS, GNSS, flash-buffered telemetry), and power subsystems.
- Architect and ship IORA AI, a RAG document-intelligence platform (FastAPI + Next.js on Railway): Supabase Postgres + pgvector retrieval with grounded-answer coverage gating, persistent per-user memory, a DuckDB text-to-SQL analytics path, and multi-provider LLM fallback (Gemini, Groq, OpenRouter).

Software Testing & Product Development Intern | IAMAI (Internet and Mobile Association of India), Delhi | May 2026 – Jul 2026
- Executed k6 load, stress, and spike testing on staging web platforms; documented vulnerability findings and consolidated performance reports, improving pre-release reliability.
- Contributed map feature development, level-of-detail (LOD) visualization, and feature validation to a GIS-based mapping platform for large-scale spatial data, strengthening spatial-data engineering skills.

Computer Vision Engineer (Intern) | Proeffico Solutions Pvt. Ltd., Noida | May 2025 – Oct 2025 (on-site May–Jul; remote Aug–Oct)
- Designed and deployed 4 production computer vision pipelines (speed monitoring, pothole detection, fire detection, workplace safety analytics) on NVIDIA Jetson Xavier NX, achieving <40 ms real-time inference latency — reducing manual monitoring effort and enabling automated real-time decisions in client workflows.

Team Lead, Rover | Technocrats Robotics, VIT Chennai | Aug 2025 – Present (member since Mar 2024)
- Lead the rover autonomy stack: CV-based soil classification and autonomous arrow-marker detection under rough terrain, deployed on Raspberry Pi and an i7 mini-PC; built an end-effector camera pipeline for tool identification and autonomous approach for manual-trigger pickup.
- All-India Rank 1, DD Robocon Round 1 (100/100); IRC 2025 — 20th worldwide out of 100+ applicant teams; IRC 2026 Finalist.

PROJECTS

SemantiSearch — Semantic Retrieval System & RAG Pipelines | MiniLM · GMM · FAISS · FastAPI · Docker
- Engineered and containerized a semantic retrieval backend as a FastAPI/Docker microservice; validated the end-to-end RAG pipeline with OpenAI at ~480 ms p95 latency with confirmed LangChain/LlamaIndex compatibility.
- Implemented GMM-based cluster-partitioned semantic caching to group similar query embeddings, cutting redundant FAISS lookups by ~40% and accelerating average query response time.

CraveConnect — Explainable Customer Intelligence System | Power BI · Gradient Boosting · SHAP · LIME · Flask
- Delivered an end-to-end customer-prediction system on 2,000 customers (19 features, 4 engineered), achieving 0.784 ROC-AUC with Gradient Boosting; served SHAP/LIME-explainable predictions via a Flask API on Render, integrated with Power BI dashboards.

Annadata — Agricultural AI Platform (SIH 2025 Finalist) | PyTorch · YOLOv8 · ResNet-34 · Flask · Python
- Built an AI agritech platform with PyTorch-based YOLOv8 and ResNet-34 models, achieving 91% disease classification and 97% soil health prediction accuracy, with GPS-enabled asynchronous pipelines sustaining ~30 FPS video-stream inference.

RESEARCH & PATENTS

Patrolling Bot — AI-Powered Autonomous Campus Surveillance Robot | Patent filed; publication pending
- Invented an autonomous patrolling robot using YOLO-based detection (missing ID cards, fire/smoke, overcrowding) and action-recognition models for fight detection, with real-time alerts to security; embedded deployment on Jetson Xavier NX / Raspberry Pi with LiDAR/ultrasonic navigation (Python, OpenCV, PyTorch, ROS).

Vision-Based Terrain Analysis | Accepted; publication forthcoming
- Developed a real-time multi-model perception system (YOLOv8 + MiDaS, PyTorch) for terrain classification in autonomous navigation; deployed to embedded edge hardware.

Few-Shot Learning with Explainable AI | Peer-reviewed journal, under review
- Designed an N-way K-shot classifier for hospital-sourced low-data medical diagnosis; achieved 93% diagnostic accuracy with LIME + SHAP attribution for full prediction interpretability.

SKILLS

Programming Languages: Python, SQL
AI/ML Frameworks: PyTorch, TensorFlow, Scikit-learn, XGBoost, HuggingFace Transformers; RAG Pipeline Design, Semantic Search, Vector Embeddings, Few-Shot Learning, Reinforcement Learning, Explainable AI (SHAP, LIME), Prompt Engineering, LLM Integration (LangChain, LlamaIndex)
Computer Vision: YOLOv8, ResNet, MiDaS, OpenCV, Object Detection, Action Recognition, Depth Estimation, Real-Time Inference, Edge Deployment (NVIDIA Jetson Xavier NX, Raspberry Pi)
Deployment & Backend: FastAPI, Flask, Docker, REST APIs, Microservices, ROS, Railway, Render, Streamlit Cloud
Databases: Supabase Postgres, pgvector, Qdrant, FAISS, DuckDB, SQL
Tools: Git, Linux, k6, Power BI (Dashboarding, DAX, Data Modeling), Pandas, NumPy

ADDITIONAL INFORMATION

ACHIEVEMENTS: Patent Filed (Patrolling Bot) · All-India Rank 1, DD Robocon Round 1 (100/100) · IRC — Top 20 Worldwide (2025), Finalist/Top 30 (2026) · SIH 2025 Finalist · Team Lead, Rover — Technocrats Robotics, VIT Chennai
SOFT SKILLS: Team Leadership · Cross-Functional Collaboration · Problem Solving · Communication · Adaptability